one image X ($= \vec{x}$), one digit c: $z_c = \vec{w}_c \cdot \vec{x} + b_c$ (728, 728)

for all digits c: $\vec{z} = \hat{w} \cdot \hat{x} + \vec{b}$ (10, 10×728, 728×10, 10)

softmax: $P(c|\vec{x}) = \frac{\exp(z_c)}{\sum_{j=0}^{9} \exp(z_j)}$ "prob that $\bar{e}$ digit is c given image $\vec{x}$"

cross-entropy loss $L = -\ln P(y|\vec{x})$, where y is $\bar{e}$ correct label

we ultimately want to obtain $\frac{\partial L}{\partial w_{kc}}$ and $\frac{\partial L}{\partial b_c}$ (there are 10 tuples of $\{w_c, b_c\}$)
first, we find $\partial L / \partial z_c$ ↑ $k \in \{1, 728\}$ denotes $\bar{e}$ pixel

$$L = -\ln P(y|\vec{x})$$

$$= -\ln\left[\frac{\exp(z_y)}{\sum_{j=0}^{9}\exp(z_j)}\right]$$

$$= -z_y + \ln\left(\sum_{j=0}^{9} \exp(z_j)\right)$$

$$\frac{\partial L}{\partial z_c} = -\frac{\partial z_y}{\partial z_c} + \frac{\partial}{\partial z_c} \underbrace{\ln\left(\sum_{j=0}^{9} e^{z_j}\right)}_{= \ln(e^{z_0} + \cdots + e^{z_c} + \cdots + e^{z_9})}$$

$$= \frac{\partial/\partial c \left(\sum_{j=0}^{9} e^{z_j}\right)}{\ln\left(\sum_{j=0}^{9} e^{z_j}\right)} - \delta_{cy}, \qquad \delta_{cy} \begin{cases} = 1, & c = y \\ = 0, & c \neq y \end{cases}$$

$$= P(y|\vec{x}) - \delta_{cy}$$

then, by chain rule, $\frac{\partial L}{\partial w_{kc}} = \frac{\partial L}{\partial z_c} \frac{\partial z_c}{\partial w_{kc}}$

$$= \left[P(y|\vec{x}) - \delta_{cy}\right] x_k$$

and, $\frac{\partial L}{\partial b_c} = \frac{\partial L}{\partial z_c} \frac{\partial z_c}{\partial b_c}$

$$= P(y|\vec{x}) - \delta_{cy}$$

$z_c = \sum_{k=1}^{728} w_{kc} x_k + b_c$

for one particular w_{kc}

$\frac{\partial z_c}{\partial w_{kc}} = x_k$ ← $\bar{e}$ brightness at that pixel

$\frac{\partial z_c}{\partial b_c} = 1$

$L = -\ln P(c|\vec{x})$, $c \in \{0, \dots, 9\}$ is written in ē context of one image
for all $N = 60000$ training samples, we define our cost fⁿ as average loss

$$C = \frac{1}{N}\sum_{i=1}^{N} L_i$$

for all images, we can define a $N \times 10$ matrix P for all probs (N ↙ # training samples = 60000)

$$P = \begin{bmatrix} P(0|\vec{x}_1) & P(1|\vec{x}_1) & \dots & P(9|\vec{x}_1) \\ P(0|\vec{x}_2) & P(1|\vec{x}_2) & \dots & P(9|\vec{x}_2) \\ \vdots & & & \\ P(0|\vec{x}_{60000}) & P(1|\vec{x}_{60000}) & \dots & P(9|\vec{x}_{60000}) \end{bmatrix}, \quad P_{ic} := P(c|\vec{x}_i)$$

we can also relabel δ_{cy} as Y_c, a vector of length 10 with the c-th digit = 1 and the rest zero. ie Y_c is a one-hot encoding

then, we can denote the correct digit corresponding to training sample i, $i \in [1, N = 60000]$, as Y_{ic}.

incorporating $\partial L/\partial w_{kc}$ and $\partial L/\partial b_c$ into our cost fⁿ, we find our expressions for average gradient across all training samples

$$\frac{\partial C}{\partial w_{kc}} = \frac{1}{N}\sum_{i=1}^{N} (P_{ic} - Y_{ic})\, x_{ik} \qquad \frac{\partial C}{\partial b_c} = \frac{1}{N}\sum_{i=1}^{N} (P_{ic} - Y_{ic})$$

after each iteration, a new set of weights and biases is computed via

$$w_{new} = w - \eta \frac{\partial C}{\partial w} \qquad b_{new} = b - \eta \frac{\partial C}{\partial b}, \quad w = (\vec{w}_0, \dots, \vec{w}_9) \{\text{vectors of length } 728$$
$$b = (b_0, \dots, b_9)$$

where ē hyperparameter η is ē learning rate.